

## TESTs - Lesson 04 - Interfaces and external calls

In the HelloWorld.sol contract, if the `_restore` function's visibility is changed from internal to external, what would likely happen when another internal function attempts to call `_restore`?

- A The call would succeed without issues.
- B The call would fail because external functions cannot be called by internal functions.**
- C The compiler would automatically adjust the visibility back to internal.

When a function like `setText(string newText)` is called in a Solidity smart contract, how does the EVM internally identify which function to execute?

- A It uses the full string name of the function, e.g., "setText".
- B It uses a unique 4-byte "function selector" derived from the function's signature.**
- C It relies on the order in which functions are declared in the contract.
- D It uses the calldata directly without any specific function identification.

When an interface is attached to a smart contract in Remix IDE, what is true regarding interface conformity?

- A The IDE performs a strict conformity check, disallowing attachment if all functions aren't implemented.
- B There is no inherent conformity check; functions are called based on matching selectors, and missing ones will revert.**
- C The contract automatically inherits missing functions from the interface.

By default, what happens if you attempt to send Ether directly to a Solidity smart contract that does NOT have a receive or fallback function?

- A The Ether is held in a temporary escrow account until a function is added.
- B The transaction will revert, as the contract cannot inherently accept Ether without a specific function.**
- C The Ether is automatically forwarded to the contract deployer.

Which special function in Solidity is executed when a call to a smart contract does not match any existing function signature?

- A The receive function.
- B The constructor.
- C The fallback function.

What is the primary difference between Solidity's receive and fallback functions?

- A receive is used for state changes, while fallback is for view functions.
- B receive is executed with empty calldata and Ether transfer, while fallback is executed if no other function matches and can optionally receive Ether and calldata.
- C fallback is for internal calls only, whereas receive is for external calls.

How does Solidity's design, particularly its verbose function declarations (e.g., specifying public, view, returns (string memory)), relate to the underlying EVM?

- A It simplifies code readability for developers new to blockchain.
- B It is a direct reflection of the EVM's low-level architecture, ensuring explicit control over how functions are handled and called on-chain.
- C It allows for automatic optimization by the compiler, reducing gas costs.

In Solidity, what is the purpose of using modifiers like onlyOwner?

- A To change the data type of function parameters.
- B To implement access control logic, restricting function execution to specific addresses or conditions.
- C To make functions more gas-efficient.

When one smart contract makes an "external call" to another smart contract, what must the calling contract know about the target contract's function?

- A The full source code of the target contract.
- B Only the 4-byte function selector (MethodID) and the ABI-encoded parameters.
- C The exact gas limit required for the target function's execution.

## Which of the following statements about Solidity interfaces is true?



**A** Interfaces in Solidity must always define a constructor.

**B** Interfaces define the function signatures that a contract must implement, but they don't contain implementation logic.

**C** A contract cannot implement more than one interface.